

Homework projects for *Dynamical systems* course, 2023/2024

1. Show that a map $(q, p) \mapsto (Q, P)$ where q, p, Q, P are in $\mathbb{R}^N$ is canonical iff it obeys the following relations

$$\{Q_i, P_j\} = \delta_{i,j} \quad \text{and} \quad \{Q_i, Q_j\} = \{P_i, P_j\} = 0.$$

for $i, j \in \{1, \dots, N\}$, with respect to the Poisson brackets defined as

$$\{f, g\} = \sum_{i=1}^N \frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i}.$$

2. Show that the set of symplectic matrices of dimension $2N \times 2N$, i.e.

$$Sp(2N) = \{F \in \mathbb{R}^{2N \times 2N} : F J F^T = J\},$$

where J is the symplectic unit, is a group. Additionally, show that a transposed symplectic matrix is symplectic.

3. Explicitly show that Hamiltonian equations (for arbitrary smooth Hamilton function $H(\vec{q}, \vec{p})$) define a canonical transformation, i.e., that the mapping

$$(\vec{q}(t), \vec{p}(t)) \rightarrow (\vec{q}(t + dt), \vec{p}(t + dt))$$

preserves the Poisson brackets. As a side-result, demonstrate the Liouville theorem, namely the conservation of the phase-space volume.

4. Derive the stroboscopic map for the kicked Harper Hamiltonian which (under certain assumptions) describes the motion of an electron in a two-dimensional periodic potential and a homogeneous magnetic field

$$H(p, q, t) = K \cos p + L \cos q \sum_{m=-\infty}^{\infty} \delta(t - m\tau).$$

Illustrate the dynamics by plotting a few phase-space portraits at various values of parameters K, L , in particular for $K < L$, $K > L$ and $K = L$, and in the ranges $K, L \ll 1$ and $K, L \sim 1$.

5. A point particle of electric charge e and mass m is bouncing elastically between walls of a one-dimensional well of width a . The particle is driven by a spatially homogeneous, periodically time-dependent (piecewise constant) electric field $E(t) \equiv E(t + \tau)$, pointing to the right $E(t) = E_0 > 0$ for a half of period $0 < t < \tau/2$ and then changing sign and pointing to the left $E(t) = -E_0 < 0$ for the second half of period $\tau/2 < t < \tau$, and so on periodically. Derive the stroboscopic map describing the motion and plot some typical phase-space portraits of dynamics, for various values of non-dimensional parameters.
6. Derive a Poincaré map for a wedge billiard, i.e. a bouncing ball in a gravity field g (pointing along $-y$ axis) and bounded by a wedge $y > y_s(x) = k|x|$. Illustrate the motion by plotting several Poincaré surface-of-section portraits of typical trajectories, for various values of a non-dimensional parameter.
7. Study a dynamical system given by the following iterated map

$$x_{n+1} = f(x_n), \quad f(x) = a(x^3 - x)$$

on a phase-space — interval $\mathcal{M} = [-1, 1]$. For $a > a_c = 3\sqrt{3}/2$ the system becomes open and the orbits can leak out. Numerically estimate the fractal dimension of the repeller, i.e. determine the ‘law of decay’: the probability $P(n)$ to stay ‘alive’ for at least n iterations, for several representative parameters $a > a_c$.

8. Study the following dissipative map on a 2D torus $\mathcal{M} = [0, 1) \times [0, 1)$

$$\begin{aligned} x' &= x + y \pmod{1}, \\ y' &= by + ax' \pmod{1}, \end{aligned}$$

where b is the dissipation rate parameter, $|b| < 1$. Numerically construct the attractor set, i.e. the set of points in phase space, visited at asymptotically long times, say for $(a, b) = (1, 0.5), (1, 0.9), (0.1, 0.9)$. Estimate also the fractal dimension of the attractor.

9. Simulate the computation of Lyapunov exponent for the standard map at various values of kicking parameter k , say for $k = 0.5, 1.0, 7.0$, and for various initial conditions. Study the convergence in time and (in)dependence of the Lyapunov exponent on the initial conditions, in particular in the chaotic domain of phase space.

10. Compute the full Lyapunov spectrum of the following 4D maps $F_j : \mathbf{x} \mapsto \mathbf{x}'$, defined over 4D torus $[0, 2\pi)^4$. For vectors $\mathbf{x} = [x_i]_{i=1}^4$, $\mathbf{x}' = [x'_i]_{i=1}^4$ and $\mathbf{y} = [y_i]_{i=1}^4$ the maps are given by

$$\mathbf{x}' = \mathbf{y} + \beta[\sin(y_3), \sin(y_4), 0, 0]^T \quad \mathbf{y} = \mathbf{M}_j \mathbf{x}, \quad (1)$$

where

$$\mathbf{M}_1 = \begin{bmatrix} 2 & -2 & -1 & 0 \\ -2 & 3 & 1 & 0 \\ -1 & 2 & 2 & 1 \\ 2 & -2 & 0 & 1 \end{bmatrix}, \quad \mathbf{M}_2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & 1 \\ -1 & -1 & -2 & 0 \end{bmatrix} \quad (2)$$

and β is a free nonlinearity parameter (typically of order 0.1). Show that the maps are symplectic, and investigate the dependence of Lyapunov exponents on β . In particular, for $\beta = 0$ you may compare to the analytical results which are possible for the linear automorphism of the torus (i.e. logarithms of eigenvalues of matrices $\mathbf{M}_j$ for $j = 1, 2$).

11. Compute dynamical entropies, the metric entropy and the topological entropy, for the following map over the unit interval $[0, 1)$,

$$x_{n+1} = f(x_n),$$

where

$$\begin{aligned} f(0 \leq x < 1/2) &= 2x \pmod{1}, \\ f(1/2 \leq x < 1) &= 4x \pmod{1}. \end{aligned}$$

12. Simulate deterministic diffusion of momentum p_n for the standard map

$$\begin{aligned} p_{n+1} &= p_n + k \sin(\varphi_n), \\ \varphi_{n+1} &= \varphi_n + p_{n+1} \pmod{2\pi}. \end{aligned}$$

Estimate numerical value of the diffusion constant

$$D = \lim_{n \rightarrow \infty} \langle (p_n - p_0)^2 \rangle / (2n)$$

in two ways: (a) via direct simulation (by definition above), or (b) in terms of dynamical correlation functions, for several values of parameter k . Can you predict dependence $D(k)$ for large k ?

13. Numerically investigate decay of correlation functions of the following discrete dynamical system defined on a unit torus $x, y \in [0, 1) \times [0, 1)$

$$\begin{aligned}x_{n+1} &= x_n + y_n \pmod{1}, \\y_{n+1} &= y_n + k(x_{n+1} - 1/2) \pmod{1},\end{aligned}$$

which is fully chaotic if the parameter k is positive. If you assume that the decay of correlations is exponential $C(t) \sim \exp(-rt)$, what is the dependence of the decay constant r on parameter k ?

14. Analyse ergodicity and mixing properties for Arnold's cat map

$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_n \\ y_n \end{pmatrix} \pmod{1}$$

defined over a unit torus $[0, 1) \times [0, 1)$. As observables you may consider the Fourier modes, such as $u_{n,m}(x, y) = \sin(2\pi nx) \sin(2\pi my)$.

15. Find systematic polynomial approximations to the universal function $g(x)$ solving the Feigenbaum equations

$$g(x) = -\hat{\alpha}g(g(-x/\hat{\alpha}))$$

together with the first Feigenbaum constant $\hat{\alpha}$, as well as the leading eigenvector and eigenvalue – the second Feigenbaum constant $\hat{\delta}$ of the linearized period-doubling operator

$$L_g\phi = \hat{\delta}\phi, \quad (L_g\phi)(x) = -\hat{\alpha} \{ \phi(g(-x/\hat{\alpha})) + g'(g(-x/\hat{\alpha}))\phi(-x/\hat{\alpha}) \}.$$

16. Make a bifurcation diagram for the following 1-dimensional iterated map over the entire real line ($\mathcal{M} = (-\infty, \infty)$)

$$x_{n+1} = x_n + a \sin(2\pi x_n),$$

as you increase the value of parameter a . Determine the critical value of $a = a_c$ where the map becomes chaotic, and estimate the diffusion constant $D(a) = \lim_{n \rightarrow \infty} (x_n - x_0)^2 / (2n)$ as a function of $a > a_c$.

17. Simulate macroscopic irreversible transport with macroscopic invertible extended multibaker map $M : (x_t, y_t) \mapsto (x_{t+1}, y_{t+1})$ defined on a *periodic* strip $(x, y) \in [0, N) \times [0, 1)$, $N \in \mathbb{Z}$ as

$$M(x, y) = \begin{cases} (\lfloor x \rfloor - 1 + 3\{x\}, y/3) & : 0 \leq \{x\} < 1/3 \\ (\lfloor x \rfloor + 3(\{x\} - 1/3), (y + 1)/3) & : 1/3 \leq \{x\} < 2/3 \\ (\lfloor x \rfloor + 1 + 3(\{x\} - 2/3), (y + 2)/3) & : 2/3 \leq \{x\} < 1 \end{cases}$$

where $\lfloor x \rfloor$ is the largest integer not greater than x and $\{x\} = x - \lfloor x \rfloor$ if the fractional part of x (and arithmetic of x -coordinates is computed (mod N)). For an integer n we call the rectangle $[n - 1, n] \times [0, 1]$ the n th cell. Consequently, the phase space is composed of N cells labeled by $1, 2, \dots, N$. A point in a cell is mapped with a probability $1/3$ to a cell to the right or the cell to the left or to the same cell. The transport between the cell resembled a random walk in 1D with diffusion constant $D = 1/3$.

By thinking of points as particle, we would like check the validity of the Fick law $j = Dc'(x)$, whereby particle current j is proportional to the gradient of concentration $c(x)$. The concentration gives the probability $c(x)dx$ to be in a narrow band $[x - dx/2, x + dx/2] \times [0, 1]$.

We test Fick law by connecting the system to a ‘chemical battery’ pushing particles between cells 1 and N : After the deterministic time step (application of the map M), we perform the following stochastic ‘battery-push’ step: If the particle is found in cell 1, $x \in [0, 1)$ it is pushed by probability p to cell N , $x \rightarrow x + N - 1$ (otherwise doing nothing), while if the particle is found in the cell N , $x \in [N - 1, N)$, it is pushed-back to cell 1, $x \rightarrow x - (N - 1)$, with probability $1 - p$ (otherwise doing nothing). The parameter $p \in [0, 1]$ controls the driving of the system, $p = 1/2$ corresponds to the equilibrium situation.

Perform a Monte-Carlo simulation of the steady state of the system. Sample a long trajectory, and compute the current through the battery $J = K(t)/t$, where $K(t)$ is the number of left-pushes minus the number of right-pushes of the battery in t time steps. At the same time, measure the density of particles in each $\rho_n = M_n(t)/t$, where $M_n(t)$ is the frequency (number of time steps) that particle is found in cell n .

On the long run, as $t \rightarrow \infty$, we will have stationary distribution of particles M_n and stationary current J , up to statistical noise. Check

for different N and M that profile of M_n^* is linear with the slope proportional to $p - \frac{1}{2}$ and that $J \propto p - \frac{1}{2}$. Calculate the diffusion coefficient from the ratio between the distribution gradient and the current. Does it match the transport coefficient D of the random-walk model $D = 1/3$?

18. Solve the previous problem using Markov chain formalism. Note that the Markov matrix, partitioning into $3N$ cells (N blocks of 3 cells) can be factorized into a deterministic and stochastic (battery) part:

$$\begin{aligned}
 P &= P_{\text{battery}} P_{\text{bulk}} \\
 P_{\text{bulk}} &= \begin{pmatrix} V_2 & V_3 & 0 & \cdots & V_1 \\ V_1 & V_2 & V_3 & \cdots & 0 \\ 0 & V_1 & V_2 & & 0 \\ \vdots & & & \ddots & \\ V_3 & 0 & \cdots & V_1 & V_2 \end{pmatrix} \\
 P_{\text{battery}} &= \begin{pmatrix} (1-p)I & 0 & 0 & \cdots & (1-p)I \\ 0 & I & 0 & \cdots & 0 \\ 0 & 0 & I & & 0 \\ \vdots & & & \ddots & \\ pI & 0 & 0 & \cdots & pI \end{pmatrix}
 \end{aligned}$$

where V_i are 3×3 matrix blocks with constant $1/3$ in row i , and zeros elsewhere, and I is a 3-dimensional unit matrix.

Compute the stationary non-equilibrium probability distribution (density profile), and perhaps also the spectral gap – relaxation time. How much can you compute analytically, or do you have to resort to numerical linear algebra?

19. Construct action-angle variables for the ellipse-billiard and discuss conditions for rational tori (periodic motion). Note that the second integral of motion is given as the product of angular momenta with respect to the both foci.
20. Use resonance overlap criterion to determine the threshold for transport of energy: A ball in a constant gravitational field bounces from the floor, which oscillates as $y_{\text{tla}}(t) = y_a \cos \omega t$ with a small amplitude y_a . Write a Poincaré map with coordinates, y_n and v_n denoting the

position of the floor at n th hit and the velocity of the ball after the n th hit, respectively, and examine its phase-space portrait.

Can the kinetic energy of the ball increase arbitrarily? Is there a KAM barrier and, if so, find it. Can the results be understood analytically? Try to approximate the local Poincaré map by the standard map and use the Chirikov resonance overlap criterion to estimate the maximal energy.

21. Variant of the previous exercise, but for a ball bouncing elastically between two plates at mean distance ℓ , where the left plate is oscillating as $x_{\text{wall}}(t) = x_a \cos \omega t$, and assume $x_a \ll \ell$.
22. Make pictures of stable and unstable manifolds, W_s and W_u , for the standard map with respect to 1 : 1 resonance (primary unstable fixed point). Tune the kick parameter k such that you obtain most clear pictures. Hint: Take a large number of points (say $\sim 10^8$) along a short stable/unstable eigenvector of the Jacobi matrix near the fixed point and then (inverse) propagate it with the map.
23. Numerically analyze the breakup of KAM tori in the standard map

$$\begin{aligned} x_{n+1} &= x_n + y_n \pmod{2\pi} \\ y_{n+1} &= y_n + k \sin x_{n+1} \pmod{2\pi} \end{aligned}$$

as a function of the winding number $\lambda = \lim_{n \rightarrow \infty} \frac{\tilde{x}_n - \tilde{x}_0}{2\pi n}$, where $\tilde{x}_n$ is the unfolded phase coordinate. Determine the value of $k^*(\lambda)$ for the breakup of KAM tori with continuous-fraction represented winding numbers $\lambda_{\text{golden}} = [1, 1, \dots]$ (golden torus), and $\lambda = [1, 2, 3, \dots]$. Hint: Construct and plot the winding number as a function of initial condition $\lambda(y) = \lambda(x_0 = 0, y_0 = y)$.

24. Derive the stability (and monodromy) matrix of an orbit in an arbitrary billiard. Work in ‘optical coordinates’: z (perpendicular distance from the reference trajectory) and θ (angle w.r.t. reference trajectory),

$$\begin{pmatrix} z' \\ \theta' \end{pmatrix} = F \begin{pmatrix} z \\ \theta \end{pmatrix}.$$

Note that $|z|, \theta \ll 1$. Show that free flight over a chord of length ℓ has a stability matrix

$$F_{\text{ff}} = \begin{pmatrix} 1 & \ell \\ 0 & 1 \end{pmatrix},$$

and a bounce from the boundary at the incidence angle ψ and with curvature radius ρ has stability

$$F_{\text{bounce}} = \begin{pmatrix} -1 & 0 \\ \frac{2}{\rho \cos \psi} & -1 \end{pmatrix}.$$

25. Compute the Lyapunov exponents in the Sinai billiard, in the shape of a unit square with a central circular obstacle of radius $\rho < 1/2$, in dependence of parameter ρ . Can you qualitatively explain the results? Draw some short periodic orbits and analytically compute their stability exponents.
26. Consider a nearly circular daisy billiard with a polar boundary function $r(\varphi) = 1 + \epsilon \cos n\varphi$ for some large n . Can you derive a resonance overlap criterion for a critical $\epsilon(n)$ where the chaos sets in?
27. Show that the problem of two point masses m_1 and m_2 moving between two parallel walls in one dimension and colliding with each other and walls is equivalent to a right triangle billiard with angles α , $\pi/2 - \alpha$ and $\pi/2$. Find the relationship between m_1/m_2 and α . Make a short simulation of the Poincare cross-section of the system and at least qualitatively investigate the ergodicity for typical angles for which α/π is an irrational number.
28. Z variacijskim računom izpelji porazdelitev v prostoru Hermitskih matrik, ki maksimizira informacijsko entropijo

$$I[p] = - \int dH p(H) \log p(H) \quad (3)$$

pri pogoju fiksne variance (vezan ekstrem) $\int dH p(H) \text{tr } H^2 = \text{const.}$ To je t.i. 'maximal ignorance principle'. Pokaži, da dobiš natanko porazdelitev *GUE* oz. *GOE* za primera, ko vzameš kompleksken hermitske oz. realne simetrične matrike.